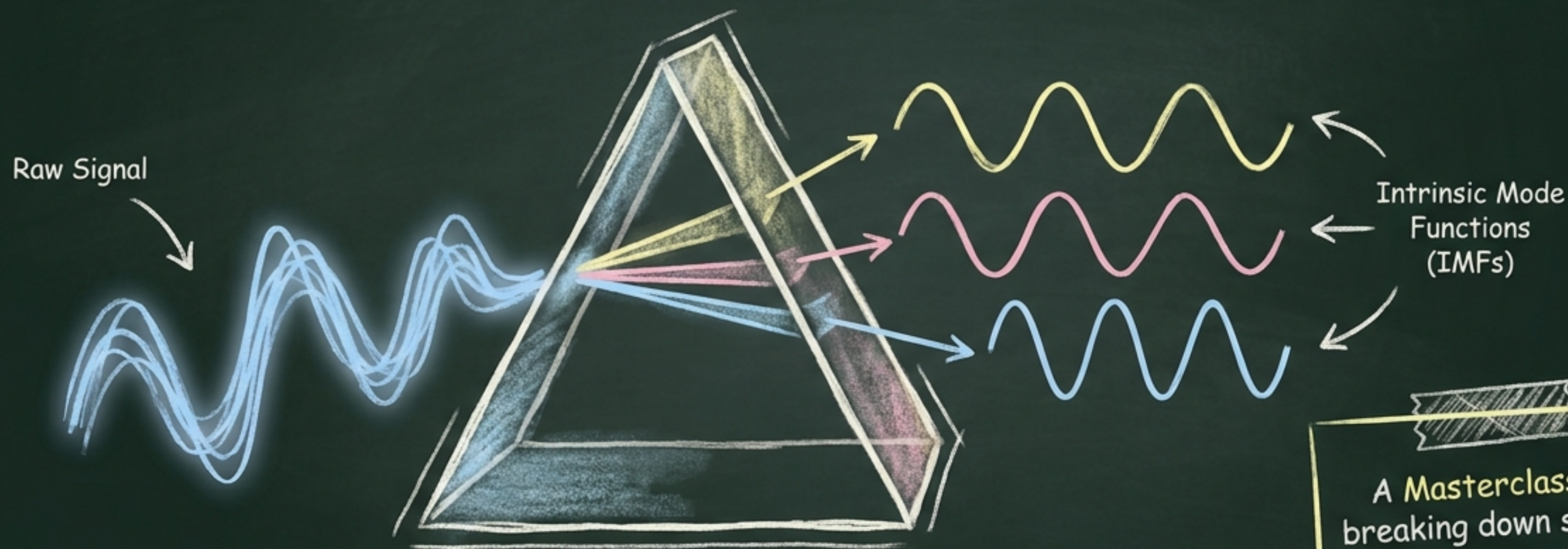


From 1D to Multivariate Physics

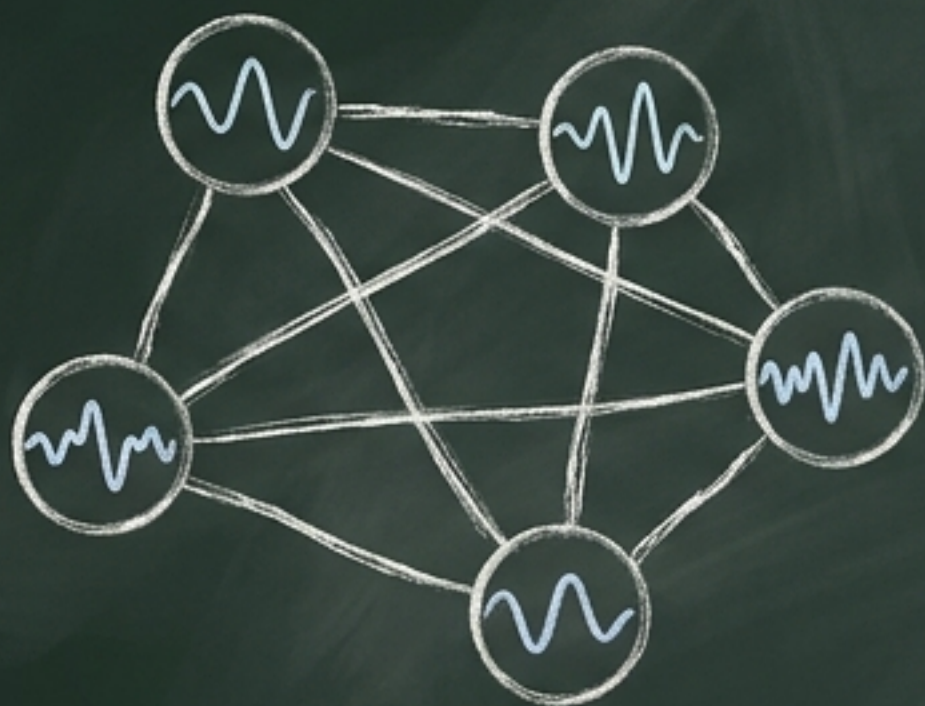
Upgrading PhaseSyncTSF (SDE12A03X) with Variational Mode Decomposition (VMD)



A **Masterclass** on breaking down single-asset time series to unlock true Kuramoto-von Mises dynamics.

The Problem with 1D Price Data

The Goal

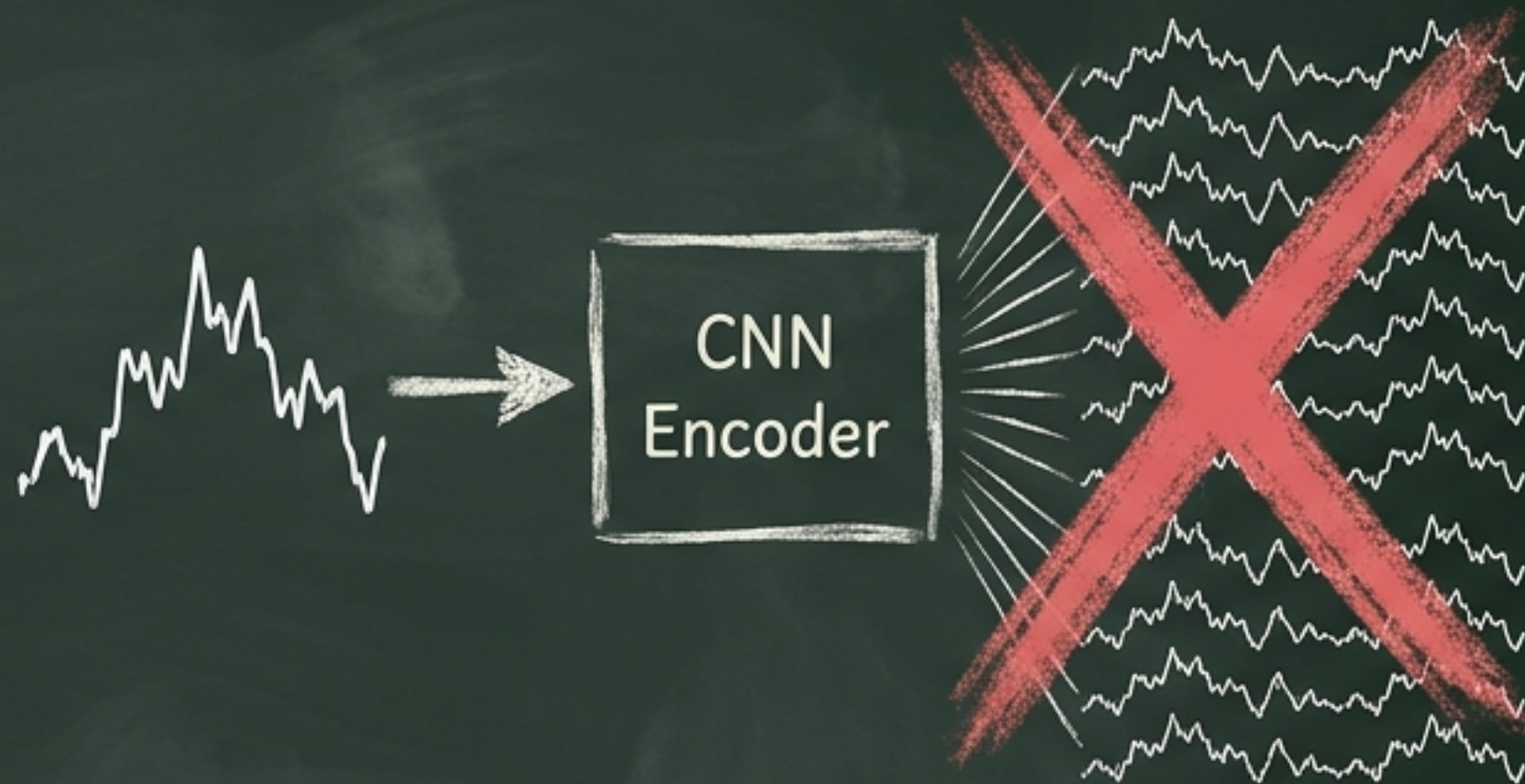


- The Kuramoto-von Mises (KvM) physics engine requires multiple independent oscillators.
- Needed for closed-form math: $\hat{k} = M^{-1}b$

Teacher's Note

We need a way to take ONE signal and mathematically shatter it into MANY independent, meaningful signals.

The Reality



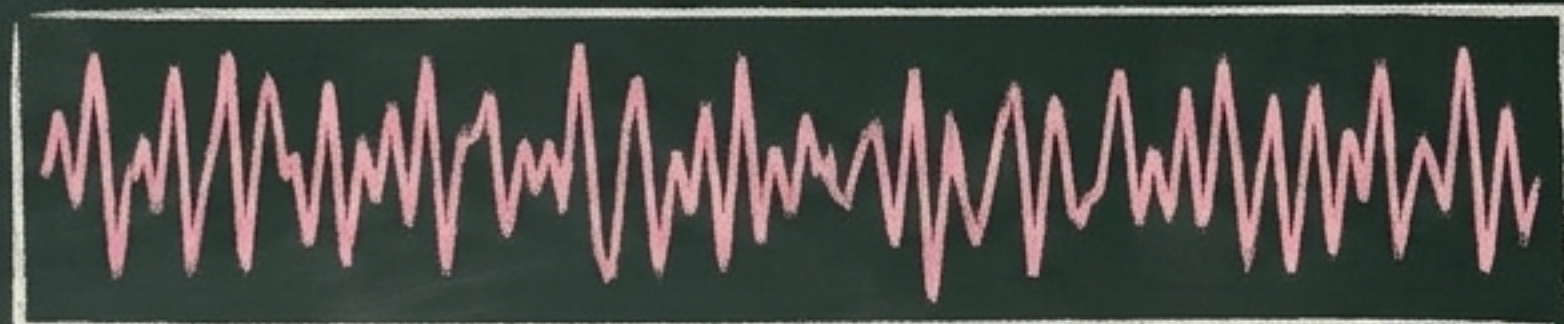
Spurious Coupling!

- Copy-pasting one 1D signal creates 100% correlated noise.
- Spatial Conditional Independence breaks.
- Physics engine learns fake relationships.

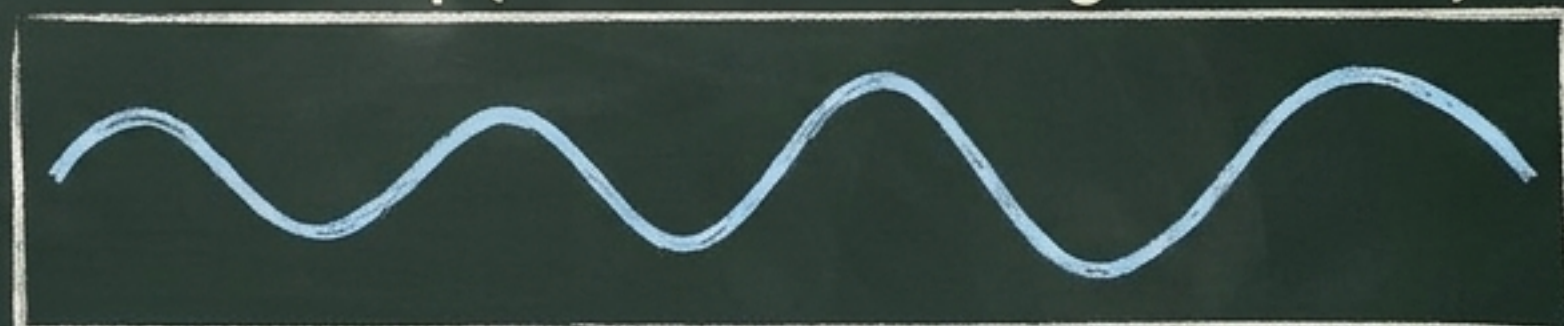
Enter Signal Decomposition



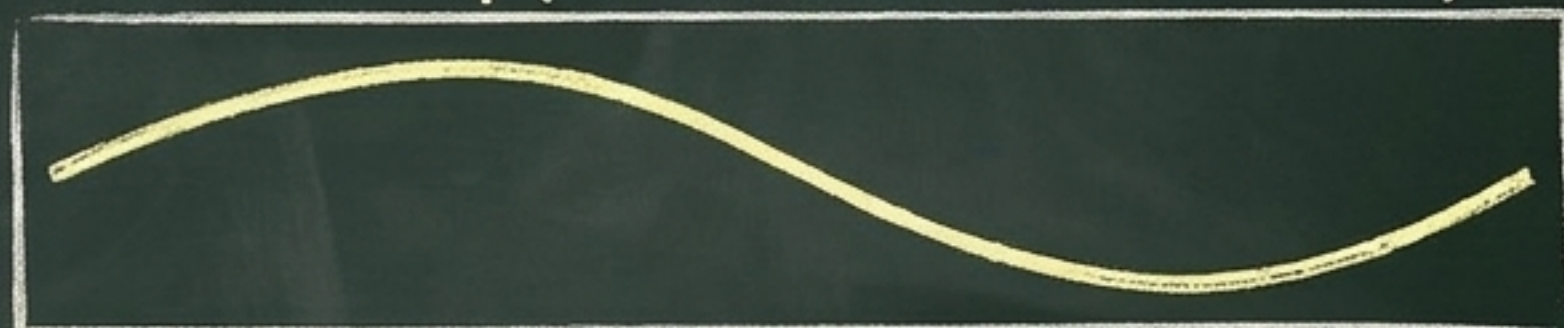
High Freq (Days - Scalpers)



Mid Freq (Weeks - Swing Traders)



Low Freq (Months - Institutions)

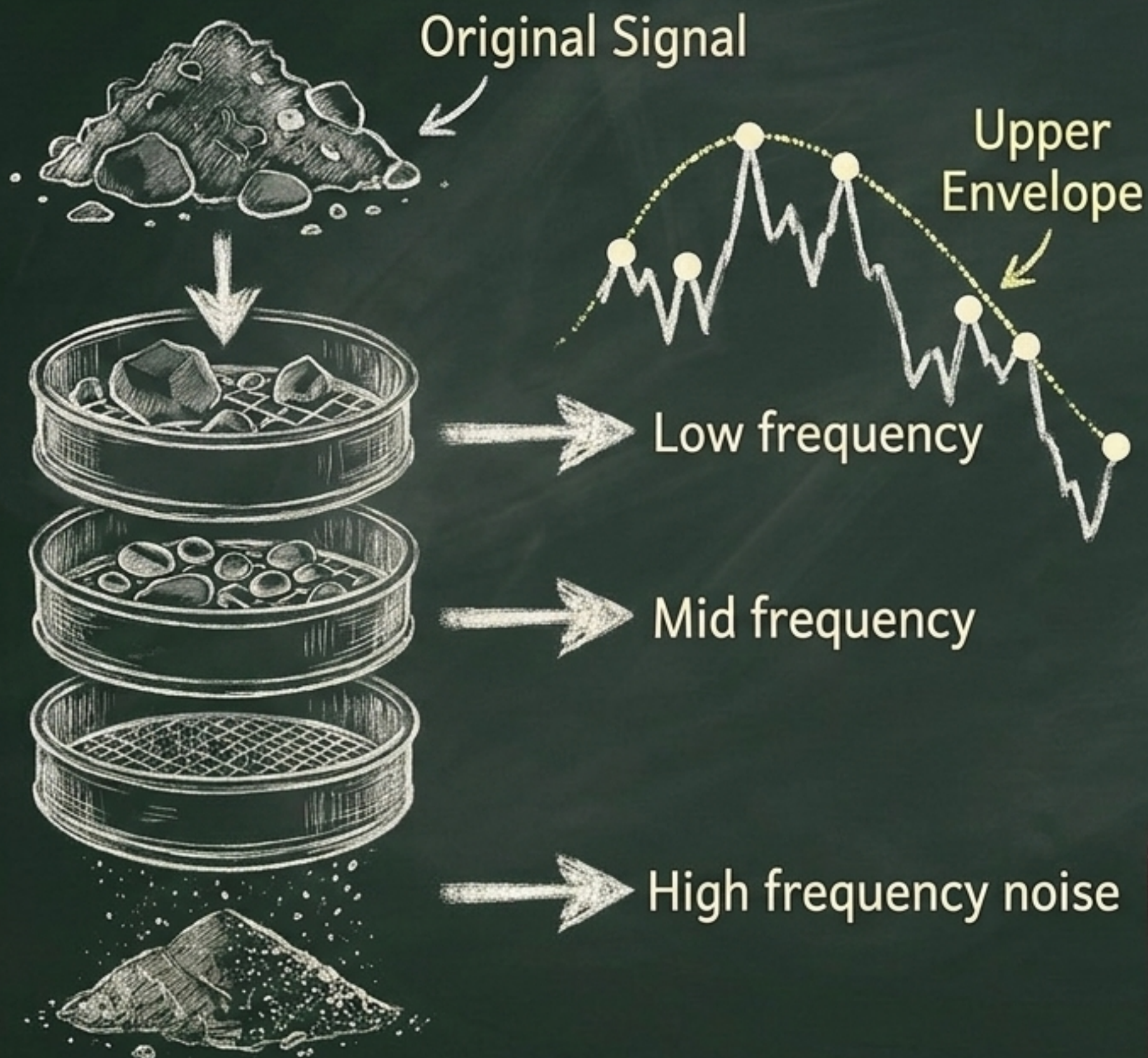


A stock price isn't a single line. It's a symphony of different traders operating at different frequencies. To create a true multivariate structure, we must extract these hidden sub-rhythms.

Two Contenders:

1. EMD (Empirical Mode Decomposition)
2. VMD (Variational Mode Decomposition)

EMD: The Empirical Sieve



How it works (Intuition):

Entirely data-driven. It finds local peaks/valleys, draws envelopes around them, subtracts the middle, and repeats. It physically sifts the signal.

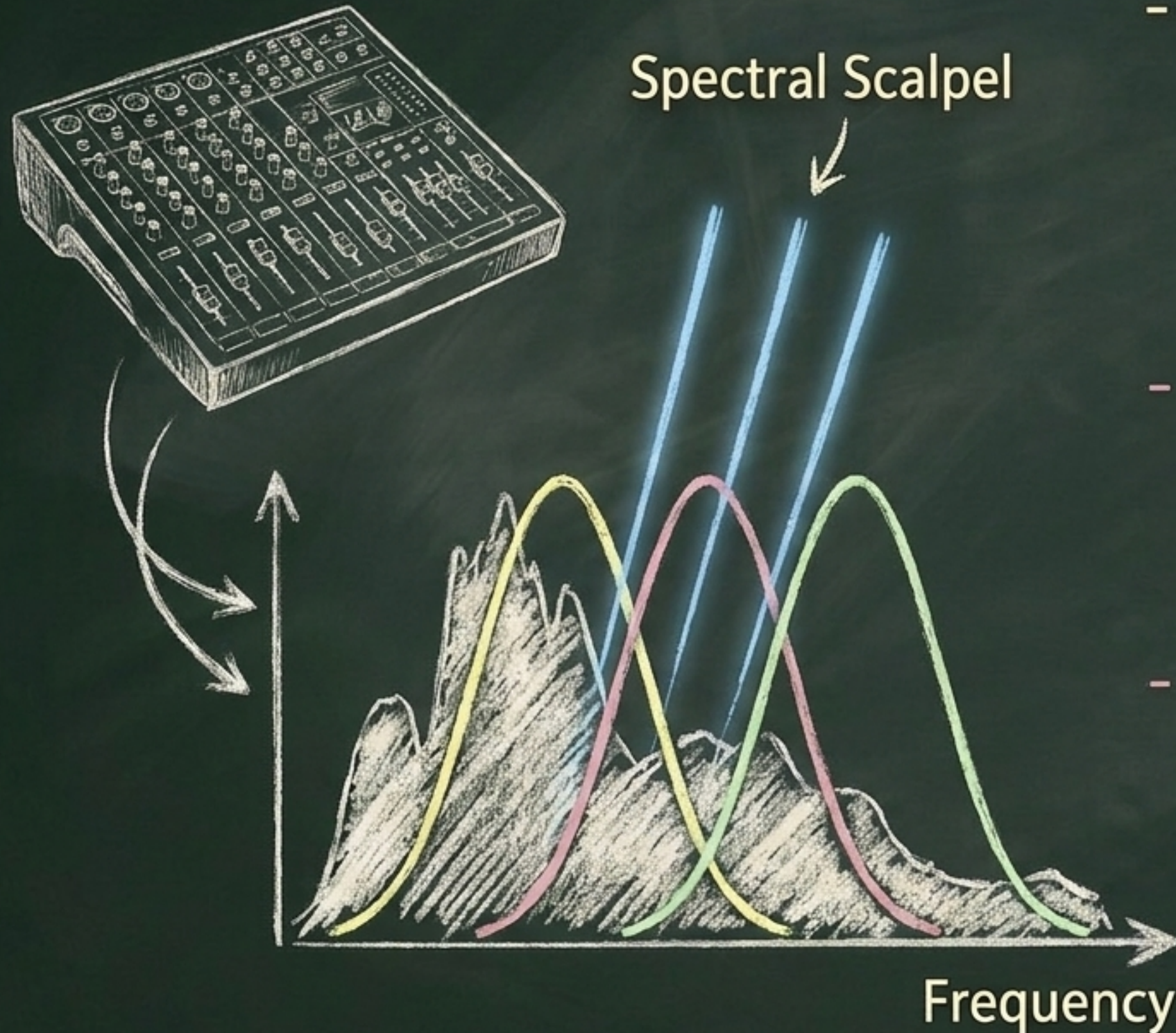
The Fatal Flaw (Mode Mixing):

Because it has no guiding math equation, a sudden market shock shatters across multiple sieves. Frequencies get hopelessly mixed up.

Teacher's Note

EMD is highly intuitive, but too fragile for our strict SDE physics engine. We need exact frequencies, not messy sand.

VMD: The Spectral Scalpel



- How it works (Intuition):

Imagine a computer scanning a symphony and declaring, "I hear exactly K different instruments." VMD optimizes all K modes simultaneously in the frequency domain.

- **The Math Goal:** Uses Variational Calculus to force each wave (IMF) to group tightly around a specific center frequency (ω_k), minimizing bandwidth.
- **The Result:** Perfect, clean, non-overlapping frequencies (Strictly Narrowband). Mode mixing is completely eliminated.

Math Note: Augmented Lagrangian and ADMM guarantee clean separation.

Why VMD Wins for PhaseSyncTSF

	EMD	VMD
Mechanism	Sequential Sifting	Simultaneous Optimization
Math Foundation	Heuristic / None	Rigorous Variational Calculus
Mode Mixing	High Risk ✖	Completely Solved ✔
Signal Output	Broadband / Messy	Strictly Narrowband ✔

The Bottom Line: The Kuramoto SDE requires Narrowband signals to extract a clean Phase via Hilbert Transform. VMD is the only tool precise enough for the job.

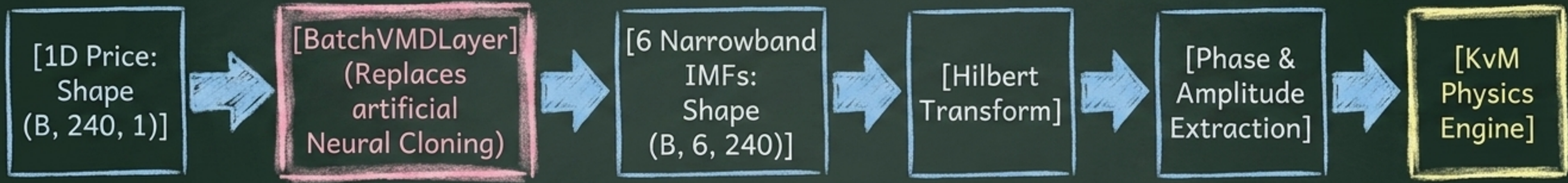
Restoring Mathematical Perfection



1. **Independence Restored:** Breaking the price into $K=6$ mathematically orthogonal Intrinsic Mode Functions (IMFs) means the noise between them is independent. Spatial Conditional Independence is saved!
2. **No Gibbs Artifacts:** Because VMD outputs pure narrowband signals, the Hilbert Transform extracts perfect instantaneous phases (θ) without edge spikes or wrapping errors.
3. **The Closed-Form Miracle:** The Stanford KvM matrix inversion now captures true market lead-lag dynamics across different time-horizons, rather than processing fake neural noise.

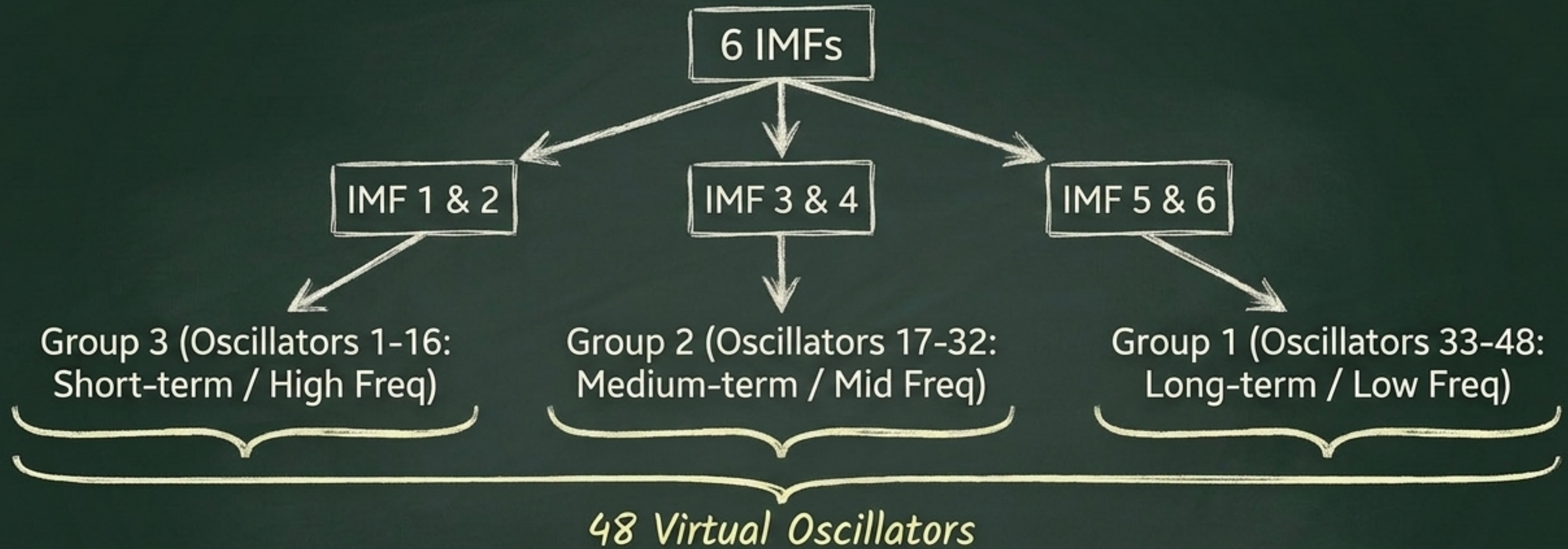
$$\hat{\mathbf{k}} = \mathbf{M}^{-1} \mathbf{b}$$

End-to-End Tensor Flow



Batch Acceleration: Implemented natively in PyTorch, the custom BatchVMDLayer processes the entire batch of 240-day lookbacks on the GPU simultaneously, ensuring no training bottlenecks.

The 1:8 Sub-Group Mapping



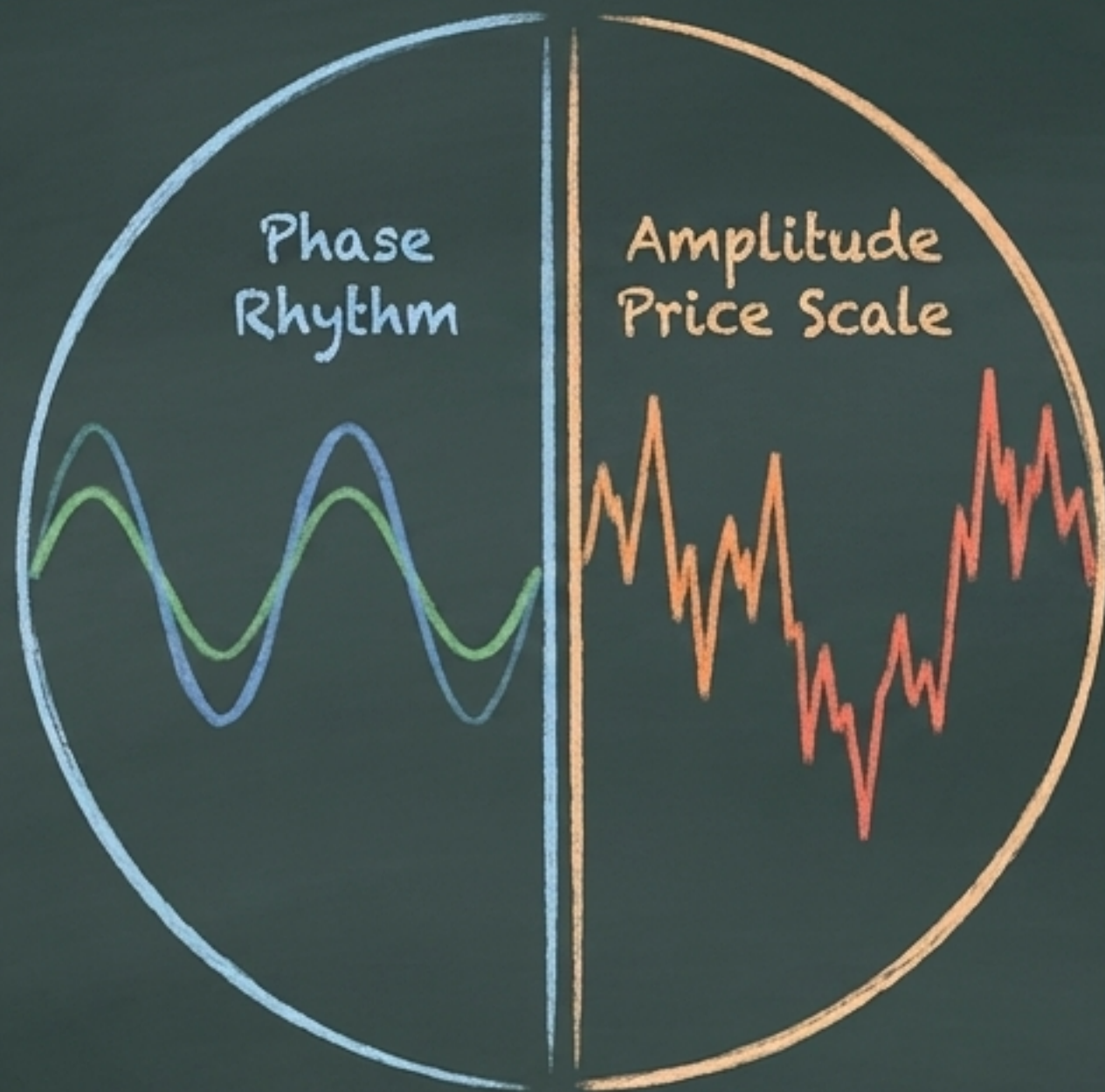
Implementation Strategy: Systematically distribute the $K=6$ mathematically pure IMFs into the 48 oscillators of the PhaseSyncTSF engine.

Physical Meaning: Instead of random latent embeddings, the dynamic coupling matrix (K_{ij}) now represents actual energy transfer and lead-lag causality between short-term market noise, weekly swings, and long-term macro trends.

The Phase-Amplitude Division of Labor

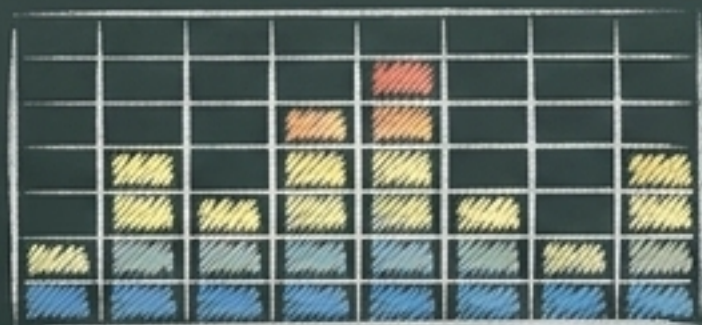
The Physics Core:

Simulates the phase rhythm forward for 24 days without losing momentum, defeating traditional AI oversmoothing.



The AI Decoder:

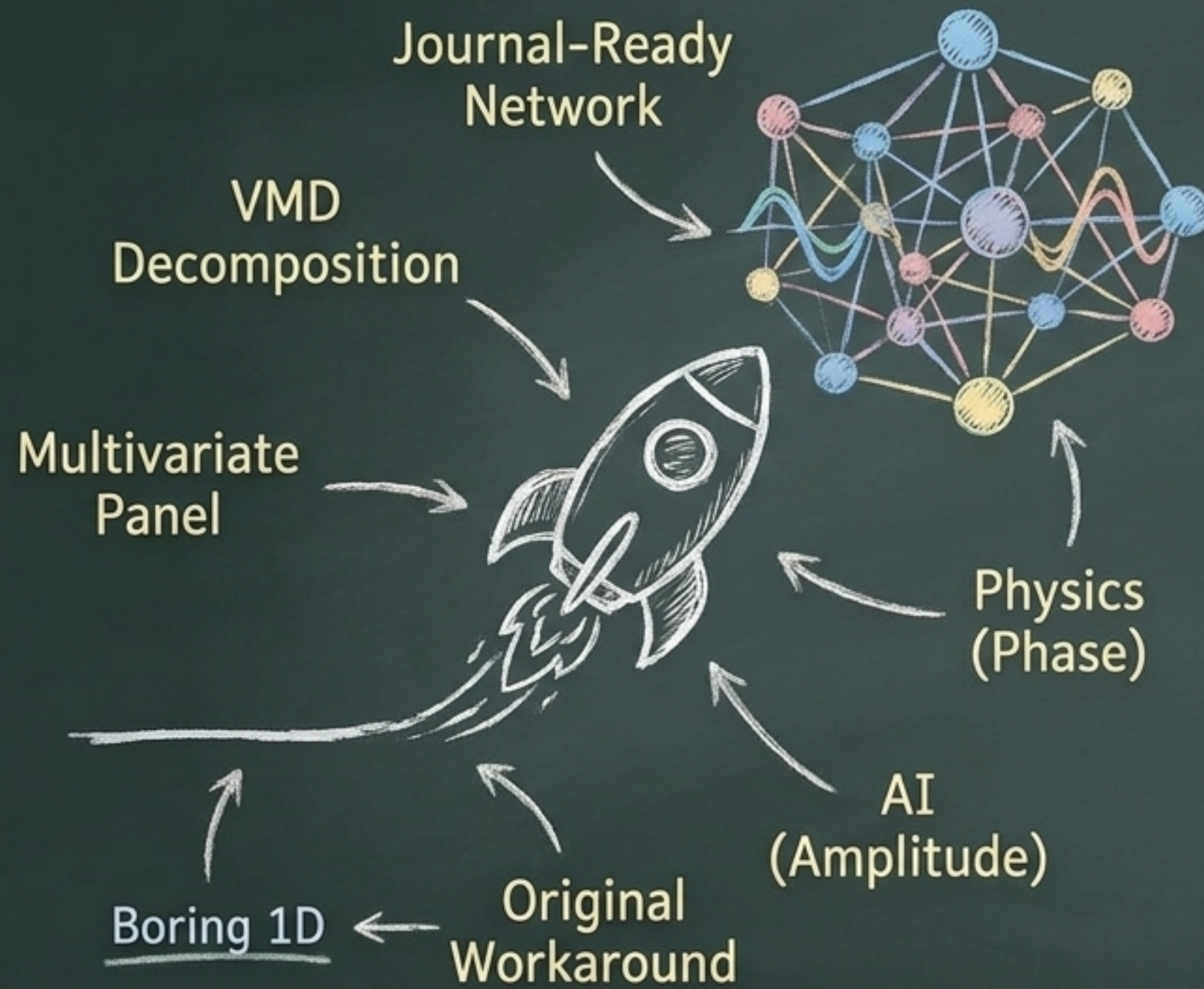
Takes the pure physical phases from the VMD-SDE pipeline and mixes them back with their original real-world price amplitudes.



RFFT Equalizer

Acts as a final high-frequency boost to ensure sharp, sudden market reversals are perfectly rendered in the final KRW/USD prediction.

From Workaround to Journal-Ready Science



• Key Takeaways:

1. EMD is intuitive but fragile; VMD is the ultimate spectral scalpel.
2. VMD converts 1D price data into a rigorous, true Multivariate panel.
3. Restoring Spatial Independence validates the Stanford KvM Closed-form MLE.
4. Physics strictly handles the Phase; AI handles the Amplitude.

Next Step to Explore:

Dynamically tuning the Heston Stochastic Volatility long-term mean ($\bar{V}$) using the real-time volatility extracted directly from our distinct VMD modes!

Thank you for joining the masterclass.